Murakkab shakllarning yuzi va perimetri

Hovuz atrofidagi yo'lka, gulzor, deraza, plitka naqshi — ko'p shakllar kvadrat, uchburchak, yarim va chorak doiralardan tuzilgan. Ularning yuzi va perimetrini bo'laklarga ajratib topamiz.

1-QISM

Mavzu

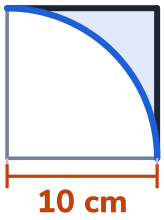
USUL · MURAKKAB SHAKLNING YUZI

1. Shaklni tanish bo'laklarga ajrating: kvadrat, to'g'ri to'rtburchak, uchburchak, doira, yarim doira, chorak doira.
2. Har bir bo'lakning yuzini toping.
3. Bo'yalgan soha bo'laklardan yig'ilgan bo'lsa — yuzlarni qo'shing. Bo'yalgan soha katta shakldan bo'lak olib tashlanib hosil bo'lgan bo'lsa — katta shakl yuzidan olib tashlangan bo'lak yuzini ayiring.

Kerakli formulalar: kvadrat $a \cdot a$; to'g'ri to'rtburchak $a \cdot b$; uchburchak $\frac{1}{2} \cdot \text{asos} \cdot \text{balandlik}$; doira πr^2; yarim doira $\frac{1}{2} \cdot \pi r^2$; chorak doira $\frac{1}{4} \cdot \pi r^2$.

1-MISOL · KVADRATDAN CHORAK DOIRA

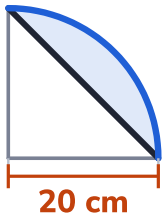
Kvadratning tomoni 10 cm. Uning ichiga radiusi 10 cm bo'lgan chorak doira chizildi. Bo'yalgan soha yuzini toping ($\pi = 3{,}14$).



1. Kvadrat: $10 \cdot 10 = 100$ cm^2.
2. Chorak doira: $\frac{1}{4} \cdot 3{,}14 \cdot 10 \cdot 10 = 78{,}5$ cm^2.
3. Bo'yalgan soha: $100 - 78{,}5 = 21{,}5$ cm^2.

2-MISOL · DOIRADAN UCHBURCHAK

Chorak doiraning radiusi 20 cm. Bo'yalgan soha yuzini toping ($\pi = 3{,}14$).



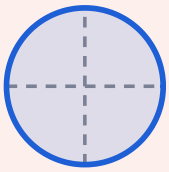
1. Chorak doira: $\frac{1}{4} \cdot 3{,}14 \cdot 20 \cdot 20 = 314$ cm^2.
2. Uchburchakning to'g'ri burchak tomonlari — radiuslar: $\frac{1}{2} \cdot 20 \cdot 20 = 200$ cm^2.
3. Bo'yalgan soha: $314 - 200 = 114$ cm^2.



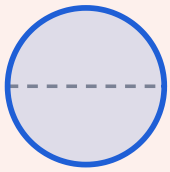
Bo'yalgan soha — kvadratning chorak doiradan ortib qolgan qismi. Shuning uchun yuzlar ayiriladi.

USUL · BO'LAKLARNI BIRLASHTIRISH

1. Radiusi bir xil bo'lgan 4 ta chorak doira birga — bitta butun doira.
2. Radiusi bir xil bo'lgan 2 ta yarim doira birga — bitta butun doira.
3. Shuning uchun bo'laklarning yuzini alohida hisoblash shart emas: bitta doiraning yuzini topish yetarli.



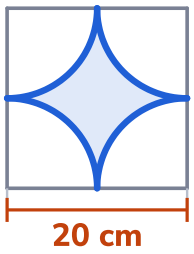
4 ta chorak doira



2 ta yarim doira

1-MISOL · BURCHAKLARDA 4 TA CHORAK DOIRA

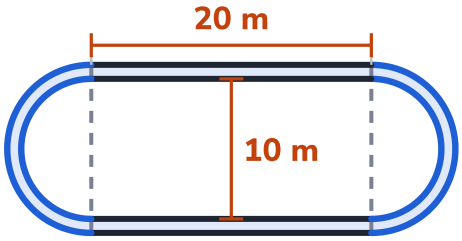
Kvadratning tomoni 20 cm. Uning har bir burchagida bir xil chorak doira chizilgan, ular bir-biriga tegib turadi. Bo'yalgan soha yuzini toping ($\pi = 3{,}14$).



1. Chorak doiralar tomonning o'rtasida tegadi: radius $20 : 2 = 10$ cm.
2. 4 ta chorak doira — bitta doira: $3{,}14 \cdot 10 \cdot 10 = 314\ \text{cm}^2$.
3. Kvadrat $20 \cdot 20 = 400\ \text{cm}^2$. Bo'yalgan soha: $400 - 314 = 86\ \text{cm}^2$.

2-MISOL · HOVUZ ATROFIDAGI YO'LKA

Hovuz to'g'ri to'rtburchak va ikki yarim doiradan iborat: to'g'ri qismining uzunligi 20 m, eni 10 m. Atrofida eni 1 m bo'lgan yo'lka bor. Yo'lkaning yuzini toping ($\pi = 3{,}14$).



1. Ikkita to'g'ri bo'lak: $2 \cdot 20 \cdot 1 = 40\ \text{m}^2$.
2. Ikki chetdagi yarim halqalar birga — bitta halqa. Radiuslar: ichki $10 : 2 = 5$ m, tashqi $5 + 1 = 6$ m.
3. Halqa: $3{,}14 \cdot 6 \cdot 6 - 3{,}14 \cdot 5 \cdot 5 = 113{,}04 - 78{,}5 = 34{,}54\ \text{m}^2$.
4. Yo'lka: $40 + 34{,}54 = 74{,}54\ \text{m}^2$.

Yuz — shakl egallagan joy, kvadrat birlikda: cm^2, m^2. Perimetr — shaklni o'rab turgan chiziq uzunligi: cm, m.

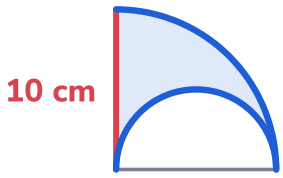


USUL · MURAKKAB SHAKLNING PERIMETRI

1. Shaklni o'rab turgan chiziqni boshidan oxirigacha aylanib chiqing. Qaysi bo'laklari yoy, qaysilari to'g'ri chiziq ekanini belgilang.
2. Har bir yoy qaysi aylananing qismi ekanini aniqlang (yarmimi, choragimi) va uzunligini toping.
3. Barcha yoy va to'g'ri chiziqlar uzunliklarini qo'shing. Shakl ichidagi chiziqlar perimetrga kirmaydi.

1-MISOL · JAVOB π ORQALI

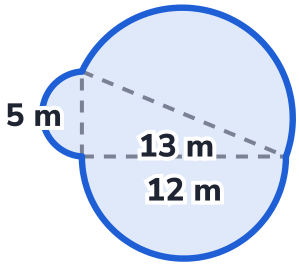
Chorak doiradan yarim doira kesib olindi. Chorak doiraning radiusi 10 cm. Bo'yalgan soha perimetrini π orqali ifodalang.



1. Katta yoy — aylananing choragi: $\frac{1}{4} \cdot 2 \cdot 10 \cdot \pi = 5\pi$ cm.
2. Kichik yoy — diametri 10 cm bo'lgan aylananing yarmi: $\frac{1}{2} \cdot 10 \cdot \pi = 5\pi$ cm.
3. Chap tomonda radius qoldi: 10 cm. Perimetr: $5\pi + 5\pi + 10 = (10\pi + 10)$ cm.

2-MISOL · UCHBURCHAK VA 3 TA YARIM DOIRA

Gulzor to'g'ri burchakli uchburchak (tomonlari 5 m, 12 m, 13 m) va uning tomonlariga chizilgan 3 ta yarim doiradan iborat. Gulzorning a) perimetrini, b) yuzini toping ($\pi = 3{,}14$).



a. Chegarada faqat 3 ta yoy bor: $\frac{1}{2} \cdot (5 + 12 + 13) \cdot \pi = 15\pi \approx 47{,}1$ m.

b. Yarim doiralar: $\frac{1}{2} \cdot 3{,}14 \cdot 84{,}5 = 132{,}665$ m^2 (bu yerda $2{,}5 \cdot 2{,}5 + 6 \cdot 6 + 6{,}5 \cdot 6{,}5 = 84{,}5$). Uchburchak: $\frac{1}{2} \cdot 5 \cdot 12 = 30$ m^2. Jami $162{,}665 \approx 162{,}67$ m^2.

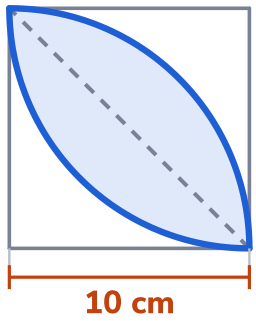
2-QISM

Misollar ishlash

1-MISOL · SHAKLNING YARMI BILAN ISHLASH

Kvadratning tomoni 10 cm. Uning ikki qarama-qarshi uchidan radiusi 10 cm bo'lgan chorak aylanalar chizildi. Bo'yalgan yaproq shaklining yuzi va perimetrini toping ($\pi = 3{,}14$).

1. Punktir diagonal yaproqni ikkita teng yarmiga ajratadi. Har bir yarmi — chorak doiradan uchburchak olib tashlangan qism (1-qismdagi kabi).
2. Yarmi: $\frac{1}{4} \cdot 3{,}14 \cdot 10 \cdot 10 - \frac{1}{2} \cdot 10 \cdot 10 = 78{,}5 - 50 = 28{,}5\ \text{cm}^2$.
3. Yaproq: $2 \cdot 28{,}5 = 57\ \text{cm}^2$.
4. Perimetr — ikkita chorak yoy: $2 \cdot \frac{1}{4} \cdot 2 \cdot 3{,}14 \cdot 10 = 31{,}4$ cm. Diagonal ichkarida — qo'shilmaydi.

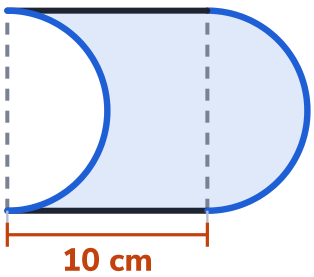


Boshqa yo'l: ikkala chorak doiraning yuzi $2 \cdot 78{,}5 = 157\ \text{cm}^2$. Ularda yaproq ikki marta sanalgan, kvadrat esa bir marta: $157 - 100 = 57\ \text{cm}^2$. Javob bir xil.

2-MISOL · BO'LAKNI KO'CHIRISH

Kvadratning tomoni 10 cm. Uning chap tomonidan yarim doira o'yib olinib, o'ng tomoniga yopishtirildi. Hosil bo'lgan shaklning yuzi va perimetrini toping ($\pi = 3{,}14$).

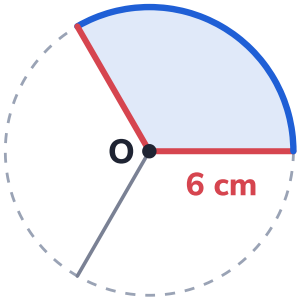
1. Qancha olingan bo'lsa, shuncha qo'shilgan. Yuz kvadratnikidek qoladi: $10 \cdot 10 = 100\ \text{cm}^2$.
2. Chegarada: tepada va pastda kvadrat tomonlari $10 + 10 = 20$ cm, chapda va o'ngda ikkita yarim aylana yoyi: $2 \cdot \frac{1}{2} \cdot 3{,}14 \cdot 10 = 31{,}4$ cm.
3. Perimetr: $20 + 31{,}4 = 51{,}4$ cm.



Yuz o'zgarmadi, perimetr esa o'zgardi: kvadratning perimetri 40 cm edi.

3-MISOL · DOIRANING $\frac{1}{3}$ QISMI

Doira 3 ta teng bo'lakka bo'lindi. Doiraning radiusi 6 cm. Bitta bo'lakning perimetri va yuzini toping ($\pi = 3{,}14$).

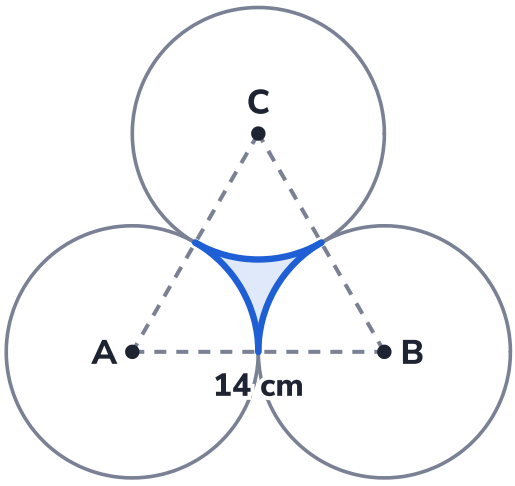


1. Bo'lakning yoyi — aylananing $\frac{1}{3}$ qismi: $\frac{1}{3} \cdot 2 \cdot 3{,}14 \cdot 6 = 12{,}56$ cm.
2. Perimetr: yoy va ikkita radius: $12{,}56 + 6 + 6 = 24{,}56$ cm.
3. Yuz — doira yuzining $\frac{1}{3}$ qismi: $\frac{1}{3} \cdot 3{,}14 \cdot 6 \cdot 6 = 37{,}68\ \text{cm}^2$.

Yarim doira — $\frac{1}{2}$, chorak doira — $\frac{1}{4}$. Bo'lak doiraning qaysi qismi bo'lsa, yoyi ham, yuzi ham shu qismga ko'paytiriladi.

4-MISOL · BIR-BIRIGA TEGIB TURGAN UCHTA DOIRA

Radiusi 7 cm bo'lgan uchta bir xil doira bir-biriga tegib turibdi. Ular orasidagi bo'yalgan soha perimetrini toping ($\pi = \frac{22}{7}$).



1. Markazlarni tutashtiramiz: $AB = BC = AC = 7 + 7 = 14$ cm. Uchburchak teng tomonli, har bir burchagi 60°.
2. 60° — to'liq aylananing $360^\circ : 60^\circ = 6$ dan bir qismi. Har bir yoy aylananing $\frac{1}{6}$ qismi, uchta yoy — $\frac{3}{6} = \frac{1}{2}$ qismi.
3. Perimetr: $\frac{1}{2} \cdot 2 \cdot \frac{22}{7} \cdot 7 = 22$ cm.

Yuzda ichki chiziq bo'laklarni ajratishga yordam beradi. Perimetrda esa u umuman sanalmaydi!



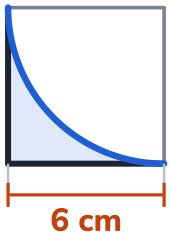
DIQQAT!

1. Perimetrga shakl ichidagi chiziq qo'shilmaydi. To'g'ri to'rtburchak (8 dm × 5 dm) va uning 8 dm li tomoni ustida yarim doira: ✗ $8+5+5+8+12{,}56$ ✓ $8+5+5+12{,}56=30{,}56$ **dm**
2. Bo'yalgan soha topilganda kattadan kichik ayiriladi, aksincha emas: ✗ $78{,}5-100$ ✓ $100-78{,}5=21{,}5\ \text{cm}^2$
3. Kvadrat burchagidagi chorak doiraning radiusi kvadrat tomoniga teng. 4 ta chorak doira tomon o'rtasida tegib tursa, radius — tomonning yarmi.
4. Ichkariga botgan yoy ham perimetrga kiradi. Yuzda esa uning bo'lagi ayiriladi.
5. Yuz cm^2, m^2 da, perimetr cm, m da yoziladi.

3-QISM

Mustahkamlash

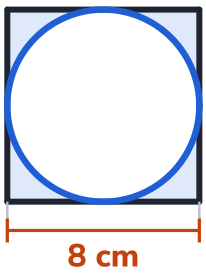
1. Bo'yalgan soha yuzini toping ($\pi=3{,}14$).



a)



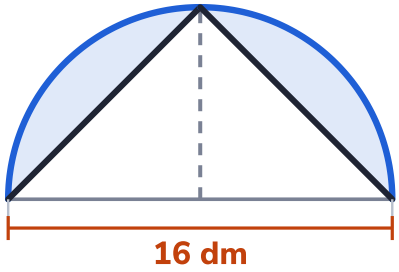
b)



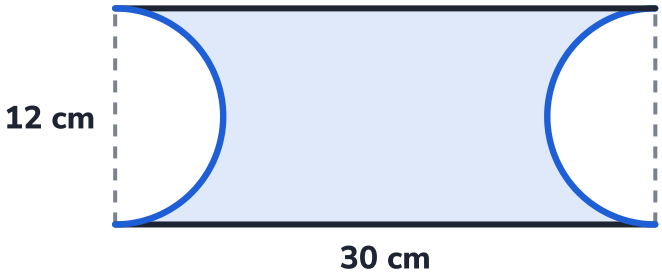
c) doira kvadratga tegib turadi

2. Kvadratning tomoni 21 cm. Uning ikki qarama-qarshi uchidan radiusi 21 cm bo'lgan chorak aylanalar chizildi. Hosil bo'lgan yaproq shaklining yuzi va perimetrini toping ($\pi = \frac{22}{7}$).

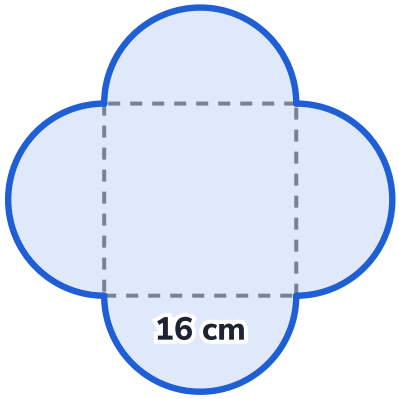
3. Yarim doiraning diametri 16 dm. Uning ichiga uchburchak chizildi: asosi — diametr, uchi — yoyning o'rtasida. Bo'yalgan soha yuzini toping ($\pi = 3{,}14$).



4. To'g'ri to'rtburchak shaklidagi taxtaning bo'yi 30 cm, eni 12 cm. Uning ikkala qisqa tomonidan diametri 12 cm bo'lgan yarim doira o'yib olindi. Qolgan qismning yuzi va perimetrini toping ($\pi = 3{,}14$).

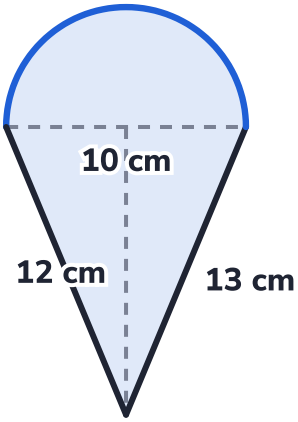


5. Plitka naqshi kvadrat va uning har bir tomoniga tashqaridan chizilgan yarim doiradan iborat. Kvadratning tomoni 16 cm. Naqshning yuzi va perimetrini toping ($\pi = 3{,}14$).



6. Dumaloq tort 6 ta teng bo'lakka kesildi. Tortning radiusi 15 cm. Bitta bo'lakning perimetri va ustki yuzini toping ($\pi = 3{,}14$).

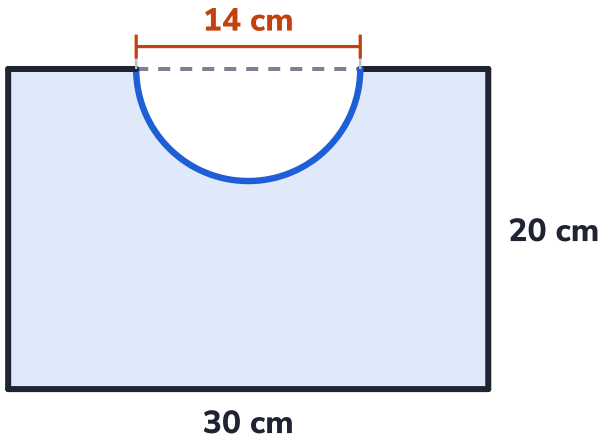
7. Muzqaymoq rasmi teng yonli uchburchak va yarim doiradan iborat. Uchburchakning yon tomonlari 13 cm, asosi 10 cm, balandligi 12 cm. Yarim doiraning diametri — uchburchak asosi. Shaklning perimetri va yuzini π orqali ifodalang.



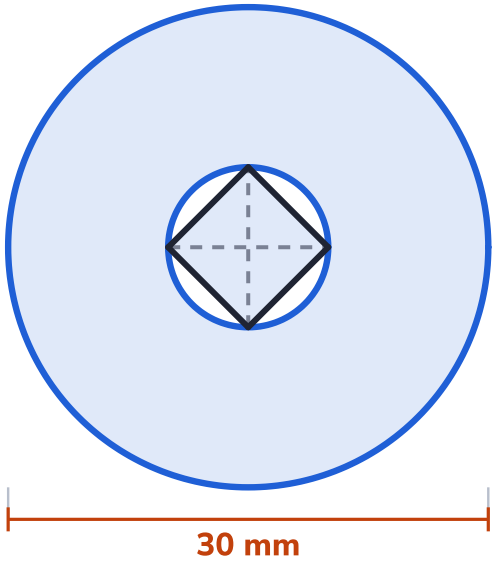
8. Sahna to'g'ri to'rtburchak va ikki yonidagi bir xil chorak doiralardan iborat. To'g'ri to'rtburchakning bo'yi 20 m, eni 7 m, chorak doiralarning radiusi ham 7 m. Sahnaning perimetri va yuzini toping ($\pi = \frac{22}{7}$).



9. Stol taxtasi to'g'ri to'rtburchak shaklida: 30 cm × 20 cm. Uzun tomonlaridan birining o'rtasidan diametri 14 cm bo'lgan yarim doira o'yib olindi. Qolgan qismning yuzi va perimetrini toping ($\pi = \frac{22}{7}$).



10. Esdalik tangasining diametri 30 mm. Uning o'rtasida diametri 10 mm bo'lgan dumaloq teshik bor. Teshik ichida kvadrat plastinka turibdi, uning uchlari teshik chetida. Bo'yalgan qismning yuzini toping ($\pi = 3{,}14$). Maslahat: kvadratning diagonallari uni 4 ta uchburchakka ajratadi.

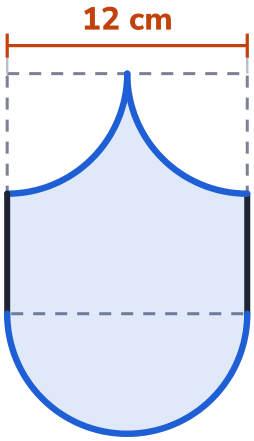


11. **Qiyinroq.** Radiusi 2 cm bo'lgan uchta bir xil tanga stolda bir-biriga tegib yotibdi. Tangalar orasida qolgan bo'sh joyning perimetrini toping ($\pi = 3{,}14$).

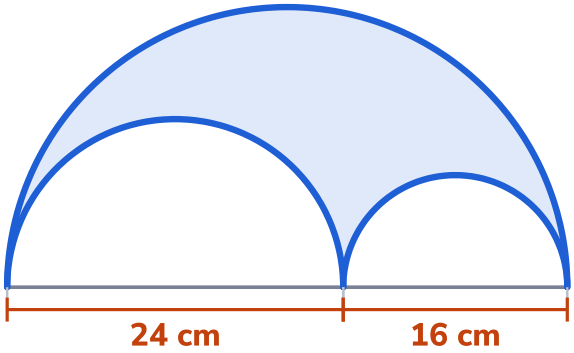
4-QISM

Mustaqil ish

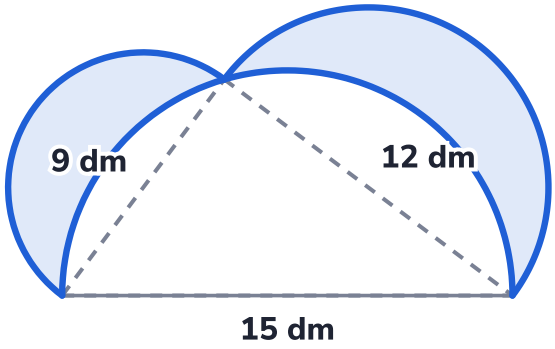
1. Nishon shakli tomoni 12 cm bo'lgan kvadratdan yasaldi: tepadagi ikki burchagidan radiusi 6 cm bo'lgan chorak doiralar kesib olindi, pastiga diametri 12 cm bo'lgan yarim doira yopishtirildi. Nishonning yuzi va perimetrini toping ($\pi = 3{,}14$).



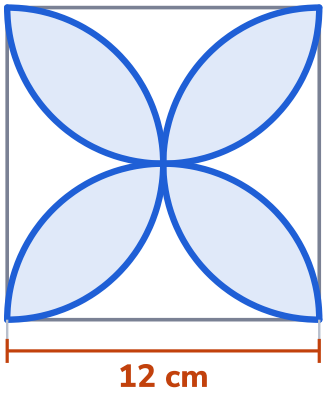
2. Katta yarim doiraning diametri 40 cm. Uning ichidan diametrlari 24 cm va 16 cm bo'lgan ikkita yarim doira kesib olindi. Bo'yalgan soha yuzini toping ($\pi = 3{,}14$).



3. **Qiyinroq.** Uchburchak to'g'ri burchakli, tomonlari 9 dm, 12 dm va 15 dm. Uning har bir tomoniga yarim doira chizildi. Bo'yalgan soha yuzini toping ($\pi = 3{,}14$). Maslahat: uchburchak va ikkita kichik yarim doiradan iborat shakl yuzidan katta yarim doira yuzini ayiring.



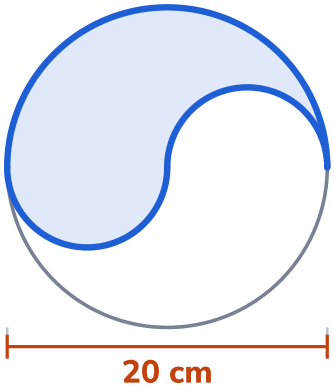
4. Kvadratning tomoni 12 cm. Uning har bir tomoniga ichkaridan yarim doira chizildi, natijada 4 ta gulbarg hosil bo'ldi. Gulbarglarning umumiy yuzi va perimetrini toping ($\pi = 3{,}14$).



5. Do'konda ikki xil dumaloq gilam sotiladi: kichigining diametri 1 m, kattasiniki 3 m.
   a) Ikkala gilamning yuzini toping ($\pi = 3{,}14$). Katta gilamning yuzi kichiginikidan necha marta katta?
   b) Katta gilam kichigidan 8 marta qimmat. Qaysi gilamning har $1\ \text{m}^2$ i arzonroq tushadi?

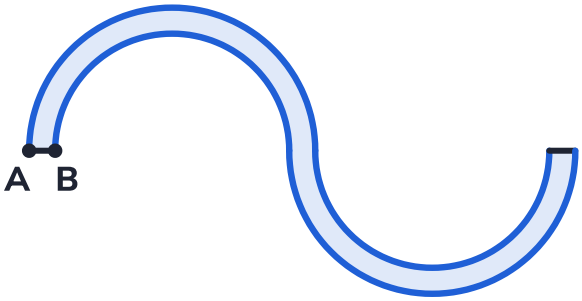
6. Ikki o'yinchoq mashina orasida 7,04 m bor. Ular bir vaqtda bir-biriga qarab yura boshladi. Ikkala mashinaning g'ildiragi ham radiusi 14 cm, har bir g'ildirak 2,5 sekundda bir marta to'liq aylanadi. Mashinalar necha sekunddan keyin uchrashadi? ($\pi = \frac{22}{7}$)

7. Doiraning diametri 20 cm. U diametri 10 cm bo'lgan ikkita kichik yarim aylana bilan ikki bo'lakka bo'lindi. Bo'yalgan bo'lakning perimetri va yuzini toping ($\pi = 3{,}14$).



8. **Xatoni toping.** Deraza to'g'ri to'rtburchak (eni 6 dm, bo'yi 4 dm) va uning ustidagi yarim doiradan iborat. Lola deraza perimetrini shunday topdi: $6 + 6 + 4 + 4 + 9{,}42 = 29{,}42$ dm. Xato nimada? To'g'ri javob qancha? ($\pi = 3{,}14$)

9. **Qiyinroq.** Bog'dagi yo'lka ikkita yarim halqadan iborat (rasm). Yo'lkaning eni 2 m. Ichki chetlarining radiusi 9 m, tashqi chetlariniki 11 m. Yo'lkaning ikkala cheti bo'ylab chiroqli sim tortildi: biri A nuqtadan, ikkinchisi B nuqtadan boshlanib, yo'lka oxirigacha boradi. Qaysi sim uzunroq? Javobni π orqali tushuntiring.



5-QISM

Javoblar

Mustahkamlash

1-masala

a) $6 \cdot 6 - \frac{1}{4} \cdot 3{,}14 \cdot 6 \cdot 6 = 36 - 28{,}26 = 7{,}74\ \text{cm}^2$; b) $10 \cdot 4 + \frac{1}{2} \cdot 3{,}14 \cdot 2 \cdot 2 = 40 + 6{,}28 = 46{,}28\ \text{cm}^2$; c) doiraning radiusi 4 cm: $8 \cdot 8 - 3{,}14 \cdot 4 \cdot 4 = 64 - 50{,}24 = 13{,}76\ \text{cm}^2$.

2-masala

Chorak doira: $\frac{1}{4} \cdot \frac{22}{7} \cdot 21 \cdot 21 = 346{,}5\ \text{cm}^2$. Yaproq: $2 \cdot 346{,}5 - 21 \cdot 21 = 693 - 441 = 252\ \text{cm}^2$. Perimetr: $2 \cdot \frac{1}{4} \cdot 2 \cdot \frac{22}{7} \cdot 21 = 66$ cm.

3-masala

Radius 8 dm. Yarim doira: $\frac{1}{2} \cdot 3{,}14 \cdot 8 \cdot 8 = 100{,}48\ \text{dm}^2$. Uchburchakning balandligi — radius: $\frac{1}{2} \cdot 16 \cdot 8 = 64\ \text{dm}^2$. Bo'yalgan soha: $100{,}48 - 64 = 36{,}48\ \text{dm}^2$.

4-masala

Ikkita yarim doira — bitta doira, radiusi 6 cm: $3{,}14 \cdot 6 \cdot 6 = 113{,}04\ \text{cm}^2$. Yuz: $30 \cdot 12 - 113{,}04 = 360 - 113{,}04 = 246{,}96\ \text{cm}^2$. Perimetr: ikkita uzun tomon $30 + 30 = 60$ cm va ikkita yarim aylana yoyi $3{,}14 \cdot 12 = 37{,}68$ cm; jami $97{,}68$ cm.

5-masala

4 ta yarim doira — 2 ta doira, radiusi 8 cm: $2 \cdot 3{,}14 \cdot 8 \cdot 8 = 401{,}92\ \text{cm}^2$. Yuz: $16 \cdot 16 + 401{,}92 = 256 + 401{,}92 = 657{,}92\ \text{cm}^2$. Perimetr — 4 ta yarim yoy, ya'ni 2 ta aylana: $2 \cdot 3{,}14 \cdot 16 = 100{,}48$ cm.

6-masala

Bo'lak — doiraning $\frac{1}{6}$ qismi. Yoy: $\frac{1}{6} \cdot 2 \cdot 3{,}14 \cdot 15 = 15{,}7$ cm; perimetr: $15{,}7 + 15 + 15 = 45{,}7$ cm. Yuz: $\frac{1}{6} \cdot 3{,}14 \cdot 15 \cdot 15 = 117{,}75\ \text{cm}^2$.

7-masala

Perimetr: $13 + 13 + \frac{1}{2} \cdot 10 \cdot \pi = (26 + 5\pi)$ cm. Yuz: uchburchak $\frac{1}{2} \cdot 10 \cdot 12 = 60\ \text{cm}^2$, yarim doira $\frac{1}{2} \cdot 5 \cdot 5 \cdot \pi = 12{,}5\pi\ \text{cm}^2$; jami $(60 + 12{,}5\pi)\ \text{cm}^2$.

8-masala

Chorak yoy: $\frac{1}{4} \cdot 2 \cdot \frac{22}{7} \cdot 7 = 11$ m. Perimetr: pastda $7 + 20 + 7 = 34$ m, tepada 20 m, ikkita yoy 22 m; jami $34 + 20 + 22 = 76$ m. Yuz: $20 \cdot 7 + 2 \cdot \frac{1}{4} \cdot \frac{22}{7} \cdot 7 \cdot 7 = 140 + 77 = 217\ \text{m}^2$.

9-masala

Yuz: $30 \cdot 20 - \frac{1}{2} \cdot \frac{22}{7} \cdot 7 \cdot 7 = 600 - 77 = 523\ \text{cm}^2$. Perimetr: $30 + 20 + 20 + (30 - 14) + \frac{1}{2} \cdot \frac{22}{7} \cdot 14 = 30 + 40 + 16 + 22 = 108$ cm.

10-masala

Tanga: $3{,}14 \cdot 15 \cdot 15 = 706{,}5\ \text{mm}^2$. Teshik: $3{,}14 \cdot 5 \cdot 5 = 78{,}5\ \text{mm}^2$. Kvadrat 4 ta uchburchakdan iborat, har biri $\frac{1}{2} \cdot 5 \cdot 5 = 12{,}5\ \text{mm}^2$: kvadrat $50\ \text{mm}^2$. Bo'yalgan qism: $706{,}5 - 78{,}5 + 50 = 678\ \text{mm}^2$.

11-masala

Markazlar orasidagi masofa $2 + 2 = 4$ cm — teng tomonli uchburchak, burchaklari 60°. Har bir yoy aylananing $\frac{1}{6}$ qismi, uchtasi — $\frac{1}{2}$ qismi: $\frac{1}{2} \cdot 2 \cdot 3{,}14 \cdot 2 = 6{,}28$ cm.

Mustaqil ish

1-masala

Ikkita kesilgan chorak doira birga — yarim doira, radiusi 6 cm; pastga ham radiusi 6 cm bo'lgan yarim doira qo'shildi. Yuz kvadratnikidek: $12 \cdot 12 = 144\ \text{cm}^2$. Perimetr: yon tomonlarda $6 + 6 = 12$ cm, ikkita chorak yoy $2 \cdot 9{,}42 = 18{,}84$ cm, pastdagi yarim yoy $\frac{1}{2} \cdot 3{,}14 \cdot 12 = 18{,}84$ cm; jami $12 + 18{,}84 + 18{,}84 = 49{,}68$ cm.

2-masala

$\frac{1}{2} \cdot 3{,}14 \cdot 20 \cdot 20 - \frac{1}{2} \cdot 3{,}14 \cdot 12 \cdot 12 - \frac{1}{2} \cdot 3{,}14 \cdot 8 \cdot 8 = 628 - 226{,}08 - 100{,}48 = 301{,}44\ \text{cm}^2$.

3-masala

Ikkita kichik yarim doira: $\frac{1}{2} \cdot 3{,}14 \cdot 4{,}5 \cdot 4{,}5 + \frac{1}{2} \cdot 3{,}14 \cdot 6 \cdot 6 = 31{,}7925 + 56{,}52 = 88{,}3125\ \text{dm}^2$. Uchburchak: $\frac{1}{2} \cdot 9 \cdot 12 = 54\ \text{dm}^2$. Katta yarim doira: $\frac{1}{2} \cdot 3{,}14 \cdot 7{,}5 \cdot 7{,}5 = 88{,}3125\ \text{dm}^2$. Bo'yalgan soha = ikkita kichik yarim doira + uchburchak − katta yarim doira $= 88{,}3125 + 54 - 88{,}3125 = 54\ \text{dm}^2$ — uchburchak yuziga teng!

4-masala

Har bir gulbarg — ikkita chorak doiradan uchburchak olib tashlangan qism (yaproq kabi), radiusi 6 cm. Gulbargning yarmi: $\frac{1}{4} \cdot 3{,}14 \cdot 6 \cdot 6 - \frac{1}{2} \cdot 6 \cdot 6 = 28{,}26 - 18 = 10{,}26\ \text{cm}^2$. 4 ta gulbarg: $8 \cdot 10{,}26 = 82{,}08\ \text{cm}^2$. Perimetr: 8 ta chorak yoy, ya'ni 2 ta aylana: $2 \cdot 2 \cdot 3{,}14 \cdot 6 = 75{,}36$ cm.

5-masala

a) Kichik gilam (radiusi 0,5 m): $3{,}14 \cdot 0{,}5 \cdot 0{,}5 = 0{,}785\ \text{m}^2$. Katta gilam (radiusi 1,5 m): $3{,}14 \cdot 1{,}5 \cdot 1{,}5 = 7{,}065\ \text{m}^2$. $7{,}065 : 0{,}785 = 9$ marta katta. b) Katta gilam 8 marta qimmat, lekin yuzi 9 marta katta. 8 ta kichik gilam narxiga katta gilam olinadi, 8 ta kichik gilam esa atigi $8 \cdot 0{,}785 = 6{,}28\ \text{m}^2$. Katta gilamning har $1\ \text{m}^2$ i arzonroq.

6-masala

G'ildirak bir aylanganda mashina aylana uzunligicha yuradi: $2 \cdot \frac{22}{7} \cdot 14 = 88$ cm. Ikkala mashina birga bir aylanishda $2 \cdot 88 = 176$ cm yaqinlashadi. $7{,}04$ m $= 704$ cm; $704 : 176 = 4$ aylanish. Vaqt: $4 \cdot 2{,}5 = 10$ sekund.

7-masala

Perimetr: katta yarim yoy $\frac{1}{2} \cdot 3{,}14 \cdot 20 = 31{,}4$ cm va ikkita kichik yarim yoy $2 \cdot \frac{1}{2} \cdot 3{,}14 \cdot 10 = 31{,}4$ cm; jami $62{,}8$ cm. Yuz: o'ngdan olingan kichik yarim doira chapga qo'shilgan, shuning uchun bo'lak — doiraning yarmi: $\frac{1}{2} \cdot 3{,}14 \cdot 10 \cdot 10 = 157\ \text{cm}^2$.

8-masala

Lola derazaning ustki tomonini (6 dm) ham qo'shdi, u yarim doira bilan to'g'ri to'rtburchak orasida — ichkarida. To'g'risi: $6 + 4 + 4 + \frac{1}{2} \cdot 3{,}14 \cdot 6 = 14 + 9{,}42 = 23{,}42$ dm.

9-masala

Har bir yarim halqada bir cheti tashqi (radiusi 11 m), ikkinchisi ichki (9 m). A dan boshlangan sim birinchi burilishda tashqi chetda, ikkinchisida ichki chetda bo'ladi. A dagi sim: $\frac{1}{2} \cdot 2 \cdot 11 \cdot \pi + \frac{1}{2} \cdot 2 \cdot 9 \cdot \pi = 11\pi + 9\pi = 20\pi$ m. B dagi sim: $9\pi + 11\pi = 20\pi$ m. Ikkala sim teng.